

Practical No. 2

Aim

Perform the Extraction Transformation and Loading (ETL) process to construct the database in the Sqlserver / Power BI.

Software Required

- Power BI Desktop
- Microsoft Excel

Dataset Used: Student Performance Dataset

The three stages of ETL are:

1. Extraction

Data is collected from one or more data sources such as Excel files, CSV files, databases or APIs.

2. Transformation

The extracted data is cleaned and modified according to analytical requirements. Common transformations include:

- Removing duplicate records
- Handling missing values
- Changing data types
- Renaming columns
- Filtering records
- Creating calculated columns
- Categorizing data

3. Loading

The transformed data is loaded into the target system or data model where it can be analyzed and visualized.

Procedure

1. Create the **Student Performance** dataset in Excel.
2. Open **Power BI** → **Get Data** → **Excel** and import the dataset.
3. Select **Transform Data** to open Power Query Editor (**Extraction**).
4. Rename columns and change the required data types.
5. Remove duplicates and blank rows.
6. Filter records with **Attendance** $\geq 75\%$.
7. Create **Total Marks** and **Average Marks** calculated columns.
8. Create a **Performance** category using conditional rules.
9. Click **Close & Apply** to load the transformed data into the Power BI data model (**Loading**).
10. Create a chart such as **Average Marks by Department** to verify the result.

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student_performance.xlsx [1]										
Sheet1										
Student_ID	Name	Department	Gender	Age	Attendance	Math	Science	English	City	
S001	Aditi	Data Science	Female	19	92	85	88	82	Mumbai	
S002	Rahul	Data Science	Male	20	78	72	75	70	Pune	
S003	Sneha	Computer Science	Female	19	95	91	89	94	Mumbai	
S004	Arjun	Computer Science	Male	21	68	65	70	62	Nashik	
S005	Priya	Data Science	Female	20	88	82	85	80	Thane	
S006	Rohan	Information Technology	Male	20	73	76	71	74	Pune	
S007	Neha	Information Technology	Female	19	91	89	92	87	Mumbai	
S008	Akash	Computer Science	Male	21	64	60	66	63	Nagpur	
S009	Isha	Data Science	Female	20	86	79	83	88	Thane	
S010	Karan	Information Technology	Male	19	81	84	78	80	Pune	

Navigator

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☒ Sheet1

Sheet1

Student_ID	Name	Department	Gender	Age	Attendai
S001	Aditi	Data Science	Female	19	
S002	Rahul	Data Science	Male	20	
S003	Sneha	Computer Science	Female	19	
S004	Arjun	Computer Science	Male	21	
S005	Priya	Data Science	Female	20	
S006	Rohan	Information Technology	Male	20	
S007	Neha	Information Technology	Female	19	
S008	Akash	Computer Science	Male	21	
S009	Isha	Data Science	Female	20	
S010	Karan	Information Technology	Male	19	

Load

Transform Data

Cancel

Student ID	Name	Department	Gender	Age	Attendance %	Math	Science	English	City	Total Marks	Average Marks	Performance Category
S001	Aditi	Data Science	Female	19	92	85	88	82	Mumbai	255	85	Excellent
S002	Rahul	Data Science	Male	20	78	72	75	70	Pune	217	72.33333333	Good
S003	Sneha	Computer Science	Female	19	95	91	89	94	Mumbai	274	91.33333333	Excellent
S004	Arjun	Computer Science	Male	21	68	65	70	62	Nashik	197	65.66666667	Need Improvement
S005	Priya	Data Science	Female	20	88	82	85	80	Thane	247	82.33333333	Excellent
S006	Rohan	Information Technology	Male	20	73	76	71	74	Pune	221	73.66666667	Good
S007	Neha	Information Technology	Female	19	91	89	92	87	Mumbai	268	89.33333333	Excellent
S008	Akash	Computer Science	Male	21	64	60	66	63	Nagpur	189	63	Need Improvement
S009	Isha	Data Science	Female	20	86	79	83	88	Thane	250	83.33333333	Excellent
S010	Karan	Information Technology	Male	19	81	84	78	80	Pune	242	80.66666667	Excellent

